Carlos Romero Madrid

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EUROPEAN MOBILITY & RESEARCH 01

01. Career & Journey

● POSICIÓN ACTUAL · CURRENT POSITION

BSC · Performance Team (HPC Software Eng.) 05/2025 – Present

- Ocean-atmosphere coupling in **DestinationE** and **ClimateDT** digital twins.
- Core development in **ECLand** climate model with Climate Change team.
- Deployment and performance profiling across **MareNostrum 5 (MN5)** & LUMI.

BSC · Fusion Team (Research Engineer) 05/2023 – 04/2025

- Advanced Computing Hub: optimized nuclear fusion simulation codes via OpenMP, MPI, PETSc, oneAPI, OpenACC on European Tier-0 supercomputers.
- Electromagnetic modeling for superconducting magnets in **ALYA-MAGNET**.

CENER · Data Scientist (Analysis & Design) 05/2022 – 09/2022

- Python pipelines for real-time wind plant telemetry, control, and predictive maintenance.

Univ. Seville & Fraunhofer FHR · Researcher 2018 – 2022

- PIC plasma simulations (Bonn); simulated annealing radar antenna design.

RIGOR & COMPUTE TOOLKIT 02

02. Education & Skills

BSc in Physics & BSc in Mathematics (Double Degree)
University of Seville (2013–2018) · **8.35/10 · 3rd of Class**

MSc in Plasma & Fusion Physics
Aix-Marseille University, France (2019–2020) · **15.43/20 · 1st of Class**

MSc in Advanced Mathematics — UNED (2023–Present, ABD)

HIGH-PERFORMANCE COMPUTING & RUNTIME STACK:

Fortran · C++ · Python · Bash · MPI · OpenMP · CUDA · OpenACC · oneAPI · PETSc · Slurm · MN5 · Leonardo · LUMI

LANGUAGES SPOKEN:

Spanish (Native) · English (C1/Prof.) · French (C1/Prof.) · Italian (Int.) · German (Int.)

PHYSICALIZATION, SOUND & VISUAL MEDIA 03

03. Creative Computing

MNusic — Real-Time Telemetry Sonification
Converts live supercomputing cluster telemetry (GPU bursts, power draw, interconnect traffic) into responsive harmonic ambient soundscapes (Web Audio API) and generative particle dynamic waves.

Pipa-Music-Ball — Bio-Kinetic Interactive Sphere
Embedded sphere with ESP32 and 6-DoF inertial sensors (accelerometer & gyro). Translates feline kinetic velocity into microtonal musical phrasing. Non-anthropocentric tangible interaction.

Expo-Curator Studio (github.com/carrommad/expo-curator)
Open-source software for automated image sub-classification, booklet publishing, and 2D/3D gallery wall staging for exhibition curators and museum science dissemination.

Lens & Motion — Photography & After Effects Lab
Architectural light framing paired with Adobe After Effects procedural motion design, particle simulations, and kinetic typography for science dissemination.

VISION & STATEMENT 04

04. Cover Letter to DAV Group

1. Scientific Storytelling & Data Democratization
Turning abstract petascale simulations into tangible, sensorial human experiences. Directly proven in *MNusic* and procedural motion essays.

2. Industrial AI & Digital Twins (GDU-HPC)
Direct match with my BSC DestinationE & ClimateDT digital twin workflows, HPC model integration, and CENER turbine predictive maintenance.

3. Creative Intelligence & Physical Viz
Tangible interactive hardware (Pipa-music-ball), participatory spatial environments, exhibition curation (Expo-Curator), and LLM-assisted workflows.

4. Urban Data Science (AIR / REALLOCATE)
Fluid and atmospheric physics grounding (MSc Plasma & Fluids) applied to urban air resilience and sustainable transport networks.

"Data is not just numbers; it is raw matter to be felt." DAV · BSC-CNS · 2026